



HR Analytics Dashboard

An end-to-end HR Analytics project built using SQL and Power BI to analyze employee demographics, workforce trends, attrition patterns, job satisfaction, and performance metrics. This dashboard helps HR teams and business leaders make data-driven workforce decisions.



Project Overview

Employee attrition and workforce management are critical challenges for organizations. This project analyzes HR data to uncover insights related to:

- Employee Attrition
- Workforce Demographics
- Department Performance
- Job Satisfaction
- Salary Distribution
- Employee Tenure
- Recruitment Trends

The goal is to transform raw HR data into actionable business insights through SQL analysis and interactive Power BI dashboards.



Business Objectives

The dashboard answers the following questions:

Workforce Overview

- How many employees are currently working?
- What is the overall attrition rate?
- What is the average employee age?
- What is the average monthly income?

Attrition Analysis

- Which departments have the highest attrition?
- Which age groups are leaving most frequently?
- Which job roles experience the highest turnover?
- Does salary impact attrition?

Employee Demographics

- Gender distribution
- Marital status distribution

- Education field analysis
- Department-wise employee count

Performance & Satisfaction

- Job satisfaction by department
- Work-life balance analysis
- Performance rating distribution
- Relationship between satisfaction and attrition



Business Problem

Employee turnover results in:

- Increased recruitment costs
- Productivity loss
- Knowledge loss
- Reduced team performance

This dashboard helps HR managers identify:

- ✓ High-risk employee groups
- ✓ Attrition drivers
- ✓ Workforce trends
- ✓ Employee engagement issues



Tech Stack

Tool	Purpose
SQL Server	Data Cleaning & Analysis
Power BI	Dashboard Development
Excel	Initial Data Exploration
GitHub	Project Documentation
DAX	KPI Calculations



Repository Structure

```
HR-Analytics-Dashboard/
├── datasets/
│   └── hr_data.csv
├── sql/
│   ├── schema.sql
│   ├── data_cleaning.sql
│   ├── exploratory_analysis.sql
│   ├── kpi_queries.sql
│   └── advanced_analysis.sql
├── powerbi/
│   └── HR_Analytics.pbix
├── screenshots/
│   ├── dashboard_overview.png
│   ├── attrition_analysis.png
│   ├── demographics.png
│   └── performance_analysis.png
├── documentation/
│   ├── data_dictionary.md
│   ├── business_requirements.md
│   └── insights.md
└── README.md
```



Dataset Information

Dataset Name

HR Employee Attrition Dataset

Number of Records

1470 Employees

Number of Columns

35 Columns

Key Columns

Column	Description
EmployeeNumber	Unique Employee ID
Age	Employee Age
Gender	Male/Female
Department	Employee Department
JobRole	Employee Role
MonthlyIncome	Monthly Salary
Attrition	Yes/No
JobSatisfaction	Satisfaction Score
WorkLifeBalance	Work-Life Rating
YearsAtCompany	Employee Tenure

 KPIs

The dashboard tracks the following KPIs:

Workforce KPIs

Total Employees

Active Employees

Attrition Count

Attrition Rate

Average Age

Average Salary

Average Tenure

Satisfaction KPIs

Average Job Satisfaction

Average Work-Life Balance

Performance Rating

Environment Satisfaction



Data Cleaning Process

The following data preparation steps were performed:

1. Missing Value Analysis

```
SELECT *  
FROM Employees  
WHERE ColumnName IS NULL;
```

2. Duplicate Detection

```
SELECT EmployeeNumber,  
       COUNT(*)  
FROM Employees  
GROUP BY EmployeeNumber  
HAVING COUNT(*) > 1;
```

3. Data Type Validation

- Age → Integer
 - Income → Numeric
 - Attrition → Categorical
-

4. Data Standardization

Examples:

```
Male  
male  
MALE
```

converted into:

```
Male
```



SQL Analysis

Attrition Rate

```
SELECT

COUNT(
CASE
WHEN Attrition = 'Yes'
THEN 1
END
) * 100.0

/

COUNT(*) AS AttritionRate

FROM Employees;
```

Department-wise Attrition

```
SELECT Department,

COUNT(
CASE
WHEN Attrition='Yes'
THEN 1
END
) AS AttritionCount

FROM Employees

GROUP BY Department;
```

Average Salary by Department

```
SELECT Department,

AVG(MonthlyIncome) AS AvgSalary

FROM Employees

GROUP BY Department;
```



Dashboard Pages

Page 1: Executive Summary

KPIs

- Total Employees
- Attrition Rate
- Average Salary
- Average Age

Visuals

- KPI Cards
 - Department Distribution
 - Gender Breakdown
-

Page 2: Attrition Analysis

Visuals

- Attrition by Department
- Attrition by Job Role
- Attrition by Age Group
- Attrition by Salary Band

Key Goal

Identify employee turnover patterns.

Page 3: Employee Demographics

Visuals

- Gender Distribution
 - Marital Status
 - Education Field
 - Department Breakdown
-

Page 4: Performance & Satisfaction

Visuals

- Job Satisfaction Analysis
- Work-Life Balance
- Performance Ratings

- Satisfaction vs Attrition

Key Insights

Example findings:

Insight 1

Sales Department shows the highest attrition rate.

Insight 2

Employees earning below average salary are more likely to leave.

Insight 3

Employees with low job satisfaction have significantly higher attrition.

Insight 4

Most employee exits occur within the first 3 years.

Dashboard Preview

Executive Dashboard

Insert Screenshot Here

Attrition Analysis

Insert Screenshot Here

Employee Demographics

Insert Screenshot Here

Performance Dashboard

Insert Screenshot Here



How to Run This Project

Step 1

Clone repository

```
git clone https://github.com/yourusername/hr-analytics-dashboard.git
```

Step 2

Import dataset into SQL Server

Step 3

Execute:

```
schema.sql  
data_cleaning.sql  
exploratory_analysis.sql
```

Step 4

Open:

```
HR_Analytics.pbix
```

in Power BI Desktop.

Step 5

Refresh data connections.



Skills Demonstrated

SQL

- SELECT
 - WHERE
 - GROUP BY
 - HAVING
 - JOINS
 - CASE Statements
 - CTEs
 - Window Functions
-

Power BI

- Data Modeling
 - DAX Measures
 - Interactive Dashboards
 - KPI Cards
 - Drill-through Reports
-

Business Analysis

- Attrition Analysis
 - Workforce Analytics
 - HR KPIs
 - Employee Segmentation
-



Learning Outcomes

Through this project, I learned:

- HR Analytics Fundamentals

- Workforce KPI Design
- Data Cleaning Techniques
- Dashboard Storytelling
- Business Insight Generation
- SQL-to-Power BI Workflow



Author

Kunal Yadav

Aspiring Data Analyst

Connect

- GitHub: <https://github.com/kunalydv-2000>
- LinkedIn: Add Your LinkedIn Profile



Project Status

Completed

Future Enhancements:

- Predictive Attrition Modeling
- Employee Churn Forecasting
- Machine Learning Integration
- Department Benchmarking